

Comparative Analysis of Conventional and Ensemble Machine Learning Techniques for Sleep Disorder Classification

Ceren KAYA*

Abstract

Computer-aided automatic diagnosis systems have become increasingly important in the early detection and management of sleep disorders, as they provide rapid analytical capabilities and assist clinicians in making more accurate and consistent decisions. In this study, two prevalent sleep disorders, apnea and insomnia, were classified using four widely utilized machine learning techniques: Naive Bayes, Support Vector Machine, Logistic Regression, and Random Forest. Each classifier offers a unique analytical perspective, thereby contributing to a more comprehensive assessment of diagnostic performance. To further enhance the robustness and reliability of classification results, three ensemble learning algorithms (AdaBoost, Bagging, and Random Subspace) are employed and integrated with base classifiers. The performance of all models was quantitatively assessed using several key evaluation metrics, including accuracy, kappa coefficient, precision and Area Under the Curve, which together provide a holistic view of classification quality. According to the experimental findings, Logistic Regression stands out as the most effective individual classifier, achieving the highest accuracy rate of 94.67%. Moreover, when Logistic Regression was combined with Bagging or Random Subspace ensemble methods, additional improvements were observed across all evaluation criteria, demonstrating the potential of ensemble-based approaches to further strengthen automated sleep disorder diagnosis.

Keywords: *Classification; disease prediction; ensemble learning; machine learning; sleep disorders.*

1. Introduction

Sleep is a vital biological function that supports overall health and well-being by allowing the body and mind to restore, repair, and maintain physiological balance. However, obtaining adequate sleep duration alone does not ensure good sleep quality. Key indicators of sleep quality include sleep latency, sleep continuity, alignment with circadian rhythms, perceived restfulness upon awakening, and the absence of daytime sleepiness [1–4].

Sleep disorders may arise from genetic predispositions, comorbid health conditions, unhealthy lifestyle habits, or unidentified causes. Consequently, determining the underlying cause of a sleep disorder requires a professional medical assessment, followed by appropriate diagnostic and therapeutic procedures. Such disorders can impair the body's repair processes, reduce growth hormone secretion, weaken the immune system, and adversely affect mental health. They may also deteriorate cognitive performance, increase emotional instability and stress levels, and disrupt metabolic balance. As a result, sleep disorders are associated with a higher risk of hypertension, depression, anxiety disorders, diabetes, obesity, and cardiovascular complications [5–8].

Among various sleep disorders, insomnia is the most common. It is characterized by difficulty initiating or maintaining sleep, frequent nighttime awakenings, reduced total sleep duration, and persistent daytime fatigue. Insomnia is often linked to underlying psychological or physiological factors. In severe cases, individuals may unintentionally fall asleep during routine activities such as driving or operating machinery which poses serious safety risks. One potential contributor to such symptoms is sleep apnea, a condition marked by repeated interruptions or shallow breathing during sleep. These episodes can occur multiple times throughout the night, resulting in morning fatigue, reduced concentration, and significant long-term health consequences if untreated [9–10].

Accurate monitoring and assessment of sleep quality are therefore essential for timely and effective treatment. Machine learning (ML) methods capable of analyzing large datasets that include parameters such as heart rate, blood pressure, sleep duration, deep sleep stages, and physical activity offer rapid, sensitive, and reliable tools for evaluating sleep patterns [11].

For instance, Kumarasamy et al. [12] examined sleep quality prediction using deep learning neural network models trained on 106 days of physical activity data collected from students using the Fitbit Charge HR™. Similarly, Ravan [13] introduced an EEG-based machine learning method for evaluating sleep quality and demonstrated that a multi-class Support Vector Machine (SVM) classifier integrated with decision tree principles yielded high classification accuracy. Liu et al. [14] extracted gait features from 59 healthy individuals using a

*Corresponding author

Kinect sensor and trained machine learning models to predict sleep quality, reporting the best performance with Gaussian Processes (correlation = 0.78). Fritz et al. [15] compared models such as Logistic Regression (LR), K-Nearest Neighbor (KNN), and Random Forest (RF) based on sleep activity and indoor air quality data, showing that these approaches can offer meaningful insights into sleep behavior. Mencar et al. [16] evaluated various machine learning algorithms using demographic and questionnaire-based data to predict the severity of obstructive sleep apnea syndrome, finding superior performance for SVM and RF. Additionally, Ramachandran and Karuppiyah [17] reviewed recent studies highlighting the importance of machine learning, deep learning, and sensor fusion in detecting sleep apnea. In another study, Alazaidah et al. [18] conducted a comparative analysis of two sleep disorder datasets to identify the most effective regression model, classification model, and learning approach across several evaluation metrics.

Considering the growing utility of ML in sleep quality assessment, the present study seeks to provide clearer and more comprehensive insights into individuals' sleep health. To achieve this, four base classifiers, Naive Bayes (NB), SVM, LR and RF, have been applied to the "Sleep Disorder Diagnosis Dataset" from Kaggle, which includes features such as sleep duration, sleep quality, blood pressure, heart rate, and stress level. Subsequently, the impact of ensemble learning techniques, including AdaBoost, Bagging, and Random Subspace (RS), on sleep disorder classification performance was examined. Experimental findings revealed that RS enhances the performance of NB, while AdaBoost improves the accuracy of SVM. Among all classifiers, LR achieves the highest accuracy rate of 94.667%, demonstrating its strong effectiveness in sleep disorder classification.

2. Materials and Methods

2.1. Dataset

"Sleep Health and Lifestyle Dataset" [19] obtained from the Kaggle repository has been used to evaluate the performance of various classifiers and ensemble learning algorithms. This dataset contains 374 instances (Sleep Apnea: 78, Insomnia: 77, None (Healthy Sleep): 219) and 13 features, representing a wide range of variables related to sleep habits, daily routines, and lifestyle factors that may affect sleep quality. There was no missing values in the dataset. The primary aim of this study was to accurately classify sleep disorders (specifically insomnia and sleep apnea) based on key features included in the dataset. All analyses and experimental procedures were conducted using both Python 3.14 and Waikato Environment for Knowledge Analysis (WEKA) 3.8.6, a comprehensive machine learning platform utilized for data preprocessing, model development, and performance evaluation. Furthermore, confusion matrices of ML classifiers were plotted in Python program.

2.2. Classifiers and Ensemble Algorithms

In this study, four fundamental classification algorithms, Naive Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), and Random Forest (RF), were employed and their classification performances were compared. To further improve the obtained results, three ensemble learning algorithms (AdaBoost, Bagging, and Random Subspace (RS)) were applied to these base classifiers, and their performances were evaluated using multiple evaluation metrics.

The model development process was structured to compare the classification performance of four selected machine learning (ML) algorithms: NB, SVM, LR, and RF. The dataset, consisting of 12 features (excluding the Person ID attribute) and 374 samples, was randomly split into training (80%) and testing (20%) subsets using a hold-out validation approach. The performance of each classifier was then evaluated individually based on accuracy, precision, kappa statistic, and ROC area success [20].

The NB classifier is a simple yet effective probabilistic model that relies on Bayes' theorem and assumes conditional independence among features [21]. SVM is a supervised learning algorithm that constructs an optimal hyperplane to separate data points from different classes with maximum margin [22]. LR builds a statistical model to describe the relationship between a dependent variable and multiple independent variables through a multivariate regression framework [23]. RF algorithm, widely used for both classification and regression tasks, generates multiple decision trees from randomly selected subsets of data and aggregates their predictions through majority voting to produce robust and reliable outputs [24].

Enhancing the performance of basic ML algorithms is essential for developing high-accuracy computer-aided diagnostic systems aimed at accelerating disease detection and treatment. Prior studies have shown that ensemble learning strategies can significantly improve the predictive capability of classification models. Therefore, this study thoroughly examined the effects of three ensemble algorithms (AdaBoost, Bagging, and RS) on the performance of four base classifiers used in sleep disorder classification [25–28].

In the AdaBoost algorithm, all samples start with equal weights during the initial training phase. After each iteration, weights are increased for misclassified samples and reduced for correctly classified ones. The final ensemble model combines the weighted predictions of all learners based on their individual accuracies.

Bootstrap Aggregating (Bagging) algorithm maintains equal selection probability for all samples but trains each model on different bootstrap subsets, each representing approximately 63.2% of the full dataset. All subsets are trained using the same base classifier, and the final prediction is obtained through weighted averaging or majority voting of the individual model outputs.

The RS algorithm differs by using randomly selected subsets of features, rather than resampling data instances, to train each classifier. The predictions of these independently trained classifiers are then combined through majority voting to produce the final decision [29].

3. Results and Discussion

This study evaluates the classification performance of ML methods to support early diagnosis of sleep disorders and to facilitate timely treatment interventions. The dataset used in this research includes 11 features: Gender, Age, Occupation, Sleep Duration, Sleep Quality, Physical Activity Level, Stress Level, BMI Category, Blood Pressure, Heart Rate, and Daily Steps. Each instance was labeled with one of three sleep disorder outcomes: “Apnea,” “Insomnia,” and “None (Healthy Sleep)”. Prior to being used as inputs to the classifier model, the features were scaled using a min–max normalization procedure. The importance of features in the related dataset was investigated by plotting a feature importance graph in Python as shown in Figure 1 using the Extra Trees (ET) classifier [30].

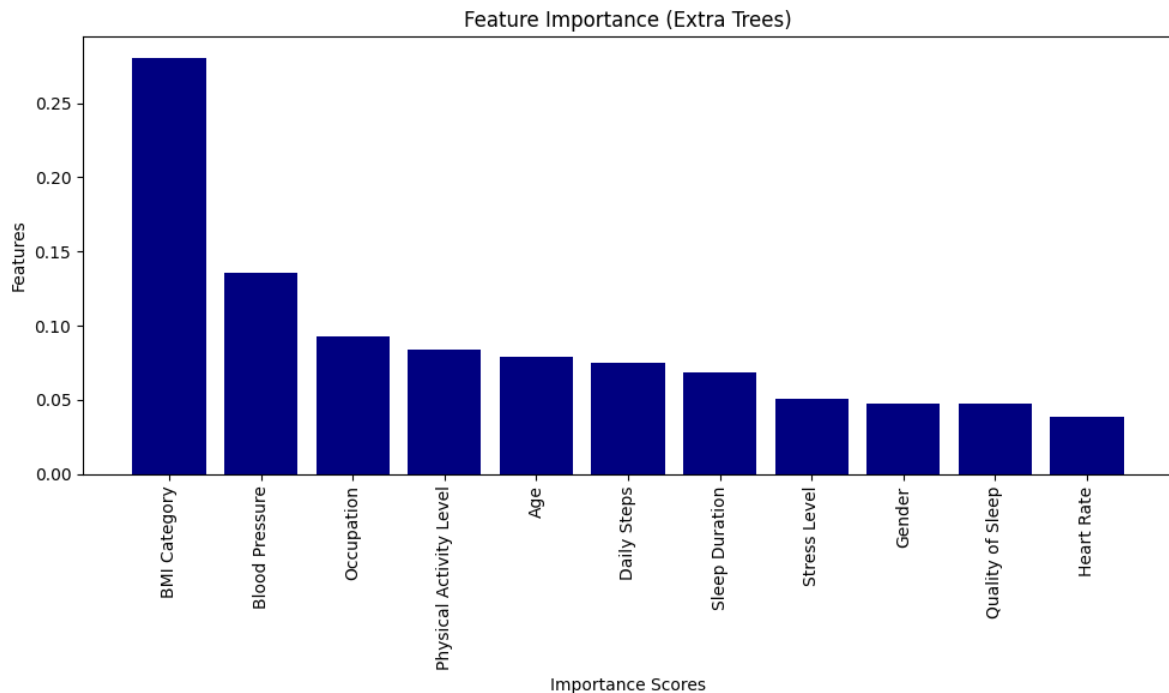


Figure 1. Feature importance of sleep health and lifestyle dataset.

From Figure 1, it has been observed that body mass index (BMI) category is the most influential feature affecting sleep quality. A healthy BMI was generally associated with better sleep, whereas abnormal BMI levels are often linked to sleep disorders. Blood pressure was the second most important factor, as normal systolic and diastolic values support healthier sleep patterns. Occupation, age, and physical activity level also contributed to differences in sleep duration, with physically demanding jobs and extreme age groups showing longer sleep tendencies. In contrast, heart rate was identified as the least impactful feature in predicting sleep disorders.

To ensure robust model evaluation, the k-fold cross-validation method was employed. In this technique, the dataset was partitioned into k subsets, where $1/k$ of the dataset was used as the test dataset and remaining portion as the training dataset in each iteration. This procedure was repeated $k = 10$ times to obtain reliable performance estimates.

The predictive performance of NB (Gaussian, Batch Size: 100), SVM (Kernel Type: Polynomial, Epsilon: $1e-12$), LR (Multinomial, Ridge Estimator: $1e-8$) and RF (Number of Estimators: 100) classifiers was assessed using accuracy, precision, kappa coefficient, and AUC metrics. Accuracy, defined in Eq. (1), represents the

proportion of correctly classified instances among the categories (Apnea, Insomnia, and None) relative to the total number of samples [31–34]. It is mathematically expressed as:

$$Accuracy = \frac{TP+TN}{TP + TN + FP + FN} \quad (1)$$

For instance, in the context of an apnea class, True Positive (TP) refers to the number of Apnea instances correctly identified as Apnea, while False Positive (FP) represents the number of non-apnea instances incorrectly classified as apnea. Similarly, False Negative (FN) denotes apnea instances that were incorrectly classified as non-apnea, and True Negative (TN) indicates non-apnea instances correctly recognized as non-apnea.

Kappa coefficient (κ) is used to quantify the degree of agreement between two classification outcomes. A κ value of 1 signifies perfect agreement, whereas a value of 0 indicates that the agreement is no better than what would be expected by random chance.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (2)$$

In this context, p_o represents the proportion of cases correctly classified by the model, while p_e denotes the probability of agreement occurring purely by chance. Kappa coefficient (κ) reaches 1 when perfect agreement is achieved and becomes 0 when the observed agreement is entirely attributable to chance.

Area Under the Curve (AUC) reflects the model's ability to distinguish between positive and negative classes, thereby indicating its overall discriminative power. To compute AUC, precision values from Eq. (3) and specificity values from Eq. (4) are utilized. An AUC value greater than 0.5 indicates that the model is able to discriminate between classes better than random guessing.

Precision measures the proportion of correctly predicted Apnea cases among all instances classified as Apnea, whereas Specificity represents the proportion of correctly identified non-Apnea cases among all actual non-Apnea instances. These evaluation metrics are calculated as follows:

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

$$Specificity = \frac{TN}{TN+FP} \quad (4)$$

Initially, the performance metrics for all classification algorithms were calculated and summarized in Table 1.

Table 1. Performance metrics of NB, SVM, LR, and RF classifiers without ensemble algorithms.

Classifier	Accuracy (%)	κ	AUC	Precision (%)
NB	90.67	0.85	0.99	92.70
SVM	90.67	0.85	0.97	93.10
LR	94.67	0.91	0.97	95.90
RF	92.00	0.87	0.99	94.40

As shown in Table 1, the LR model exhibited the highest overall classification performance. Utilizing all 11 attributes for sleep disorder classification, LR achieved an accuracy of 94.67%, a precision of 95.90%, a kappa coefficient of 0.91, and an AUC of 0.97. Although the overall accuracies of NB and RF algorithms were relatively lower, these models achieved the highest AUC values. Furthermore, the impact of ensemble algorithms on classification performance was evaluated by combining them with the base classifiers, and the resulting performance metrics were presented in Table 2.

Table 2. Performance metrics of NB, SVM, LR, and RF classifiers with Adaboost, Bagging and Random SubSpace ensemble algorithms.

Ensemble Method	Classifier	Accuracy (%)	κ	AUC	Precision (%)
Adaboost	NB	90.67	0.85	0.99	92.70
	SVM	90.67	0.85	0.94	93.10
	LR	94.67	0.91	0.97	95.90
	RF	92.00	0.87	0.96	94.40
Bagging	NB	90.67	0.85	0.99	92.70
	SVM	93.33	0.89	0.99	94.50
	LR	94.67	0.91	0.99	95.90
	RF	92.00	0.87	0.99	94.40
Random SubSpace	NB	92.00	0.87	0.99	94.40
	SVM	92.00	0.87	0.99	93.80
	LR	94.67	0.91	0.99	95.40
	RF	92.00	0.87	0.99	94.40

As shown in Table 2, all ensemble algorithms generally improved the AUC metric. Notably, the Random Subspace (RS) method increased the accuracy of NB and SVM classifiers from 90% to 92% and also positively impacted other performance measures. Bagging technique further enhanced the SVM classifier's accuracy to 93.33%, whereas AdaBoost did not produce any significant improvement in either accuracy or kappa coefficient. Overall, the LR classifier achieved the highest classification accuracy of 94.67%, independent of the ensemble application. However, when additional evaluation metrics were considered, the LR classifier combined with Bagging or RS ensembles demonstrated consistently strong and balanced performance across all criteria.

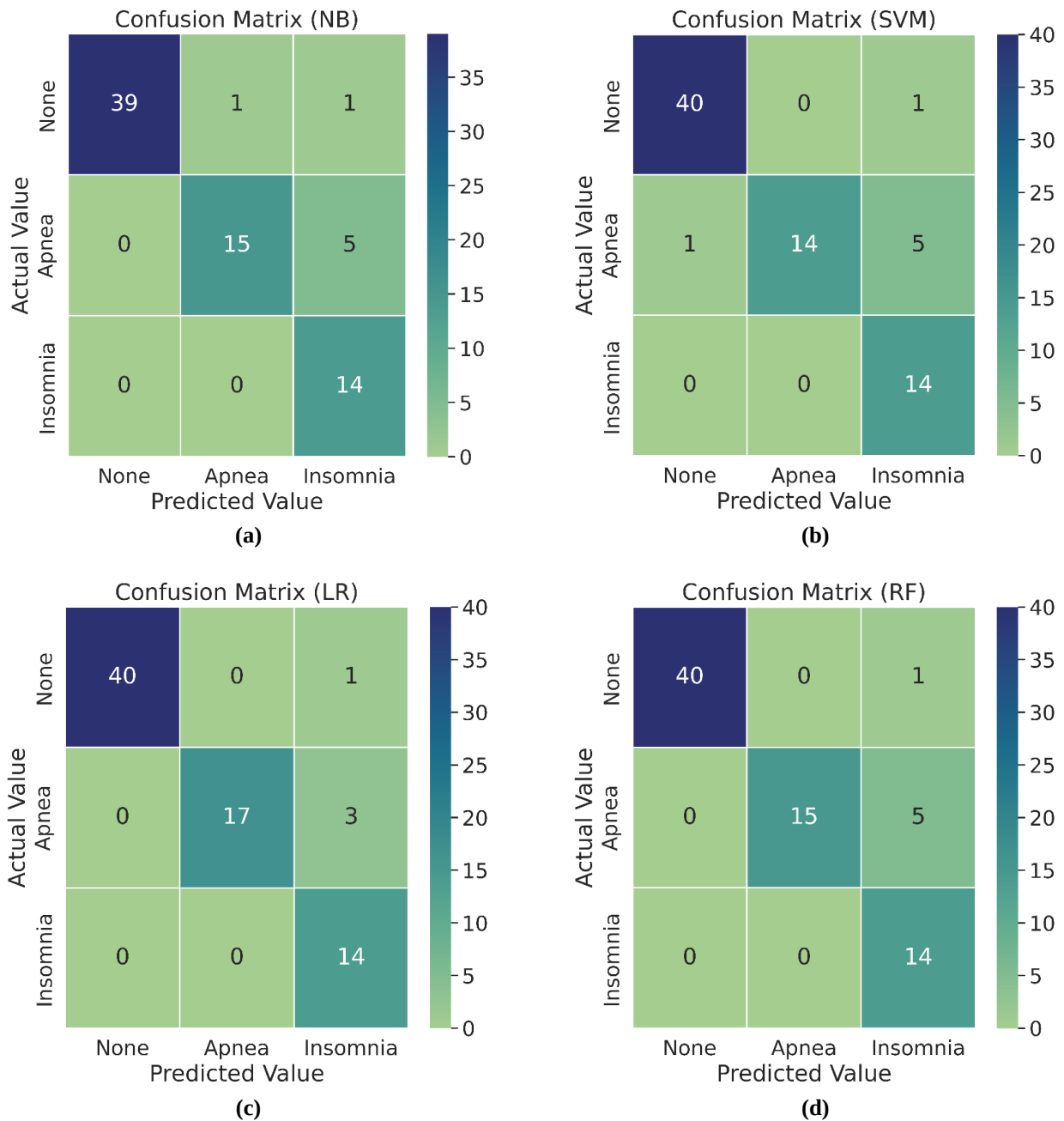


Figure 2. Confusion matrices of ML classifiers, a) NB b) SVM c) LR d) RF.

Confusion matrices of ML classifiers without and with ensemble algorithms plotted in Python were illustrated in Figure 2 and Figure 3, respectively.

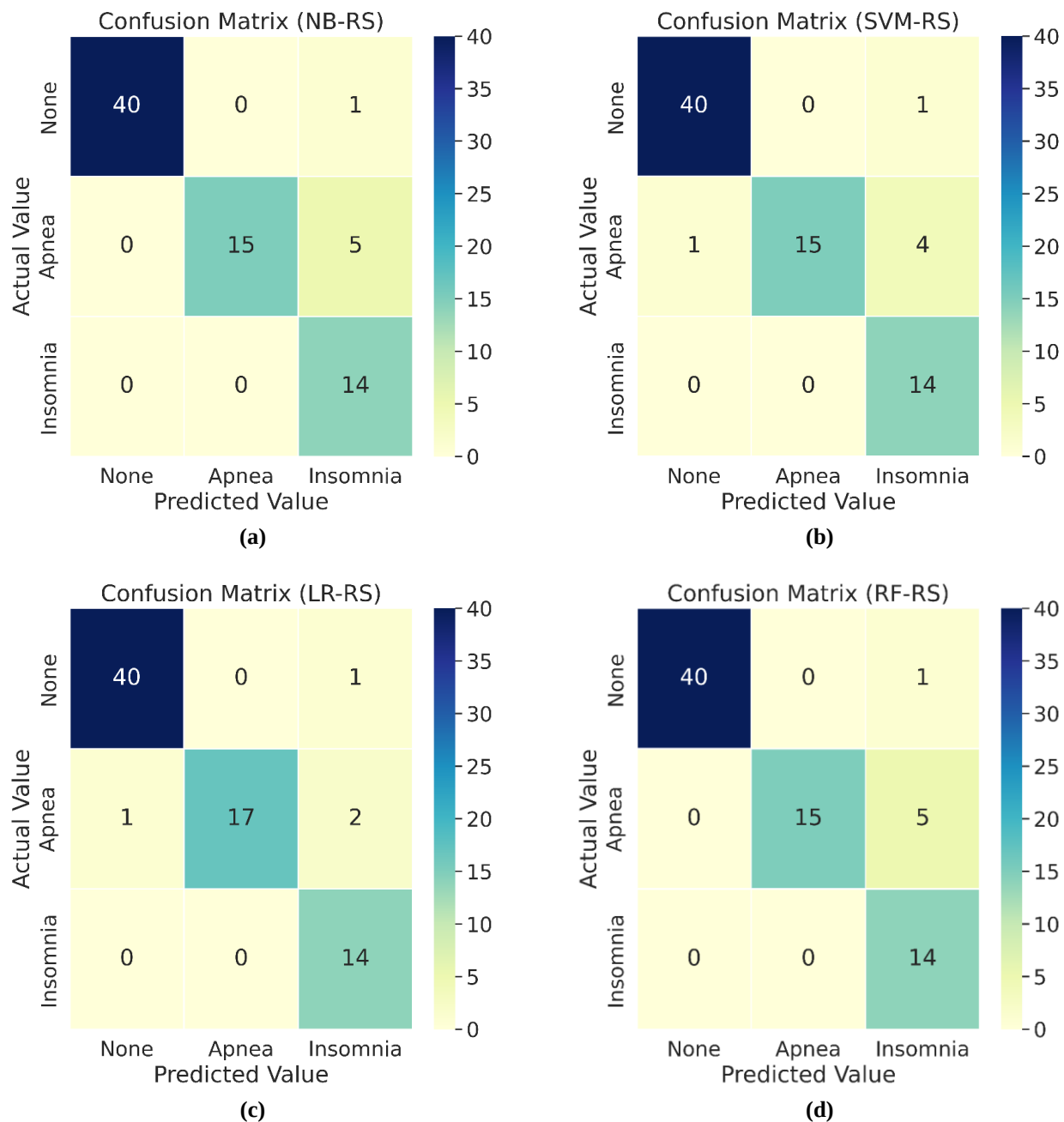


Figure 3. Confusion matrices of ML classifiers with ensemble algorithms, a) NB-RS b) SVM-RS c) LR-RS d) RF-RS.

The classification accuracy performance with previous studies conducted on the same dataset was compared in Table 3. The LR classifier achieved an outstanding 94.67% accuracy on the Sleep Health and Lifestyle Dataset, surpassing studies using the same dataset in the literature.

Table 3. Classification accuracy comparison on the same dataset.

Author(s)	Dataset	Classifiers	Accuracy (%)
Tareq [35]	Sleep Health and Lifestyle Dataset	Random Forest	88
Warunlawan et al. [36]	Sleep Health and Lifestyle Dataset	Bagged Trees	91.90
Alshammari [37]	Sleep Health and Lifestyle Dataset	Artificial Neural Network	92.92
Taher and Ayon [38]	Sleep Health and Lifestyle Dataset	Gradient Boosting	93.80
Proposed Study	Sleep Health and Lifestyle Dataset	Logistic Regression	94.67

4. Conclusion

This study aims to develop a clinical decision support system for early detection of sleep disorders, specifically apnea and insomnia, by leveraging various machine learning classifiers. In this framework, performances of four selected classifiers, Naive Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), and Random Forest (RF), were systematically evaluated using multiple analytical approaches.

Initially, the classification performance of four base classifiers was assessed using accuracy, precision, kappa coefficient, and Area Under the Curve (AUC) metrics. Subsequently, the same analyses were conducted for classifiers augmented with ensemble learning algorithms, including AdaBoost, Bagging, and Random Subspace (RS).

The results demonstrated that the LR classifier achieved the highest overall classification performance. While ensemble methods generally improved the AUC metric, LR combined with Bagging or RS ensembles delivered the most balanced and superior results across all evaluation criteria. This strong performance stems from structural properties of the feature space such as approximate linear separability, controlled multicollinearity, and stable distributions which align closely with the statistical assumptions of LR.

Consequently, LR yields stable, low-variance decision boundaries that surpass the consistency of more complex models prone to overfitting or noise sensitivity. This robustness enables LR to capture clinically meaningful patterns with high precision, making it the most strategic choice for early detection of sleep disorders and achieving a classification accuracy of 94.67%.

Declaration of Interest

The author declares that there is no conflict of interest.

Author Contributions

All aspects of this manuscript, including its conception, design, execution, data analysis, interpretation, and writing, were solely undertaken by the author.

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